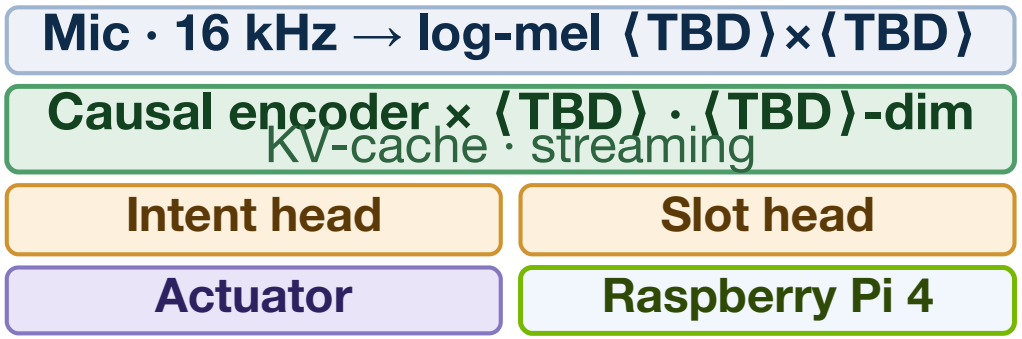
ME2 — Voice Controlled Smart Device

Always-on keyword + intent recognition at the edge · no cloud round-trip

1 October 2026 · Leandro Soneja - AI 231 ME2 · Model · Dataset · Training

Model



Item	Value
Parameters / weights	0.34 M · 0.70 MB

Dataset

Item	Value
Source	ME2 holdout (HF: airimonda/ai
Hours / utts	0.77 h / 424
Speakers	6 real + synth + 1 owner
Labels	19 intents · 3 (G3.5) slots

Training on the A100 cluster

Item	Value
Cluster	DGX Station · 1 GPU (local)
Objective	CNN (DS-CNN) classifier
Optimiser	n/a (ship model_src)
Steps / loss	n/a / n/a

Validation on the Raspberry Pi 4

Item	Value
Keyword / intent acc	91.2 % / 91.7 %
False-accept rate	100.0 %
Latency p95 / RTF	47.9 ms / 0.048
Runtime	tflite (XNNPack)

Causal encoder × n/a · n/a-dim

Mic · 16 kHz → log-mel 148×60

To Be Submitted

GitHub repositorygithub.com/airimonda/vcm-benchmark
Dataset locationhuggingface.co/datasets/airimonda/ai231-me2-voice-com
A100 cluster DGX Station - local (see ships)
Model weights Shipping/ - TFLite f16 + SHA256SUMS

What the reviewer will look for

#	Item	Status
1	Repo public, one-command reproduction	done: repo public + benchmark
2	Dataset licensed and citable (DOI)	done: LICENSE.md (research-only)
3	Training logs + final checkpoint committed	done: ships + SHA256 + mod
4	Pi 4 latency reproduced by the posted script	pending: Pi offline 2026-10-
5	Held-out test set, unseen speakers	done: holdout unseen speakers
6	Baseline of comparable size compared	done: 55-model baseline sweep

Best Combination - ME2 full-pipeline batch (2026-10-24)

424 takes x 13 Gate 1 x 55 Gate 2 x raw x arch v1+v2 | winner: gate1v3_f32 x gate2v64p_paired_owner x raw x v1: full 187/205 (91.2%), action 91.7%, wake 94.6%, OOS reject 0.0%, p95 47.9 ms

Demo setup (Pipeline test window):

Gate 1 variant = gate1v3_f32 Gate 2 model = gate2v64p_paired_owner Architecture = v1 NC = raw

Top combinations (ranked by fully-correct rate):

#	G1	G2	nc	arch	full	rate	act	wake	oosrej	ms	p95
1	v3_f32	v64p_paired_owner	raw	v1	187/205	91.2%	91.7%	94.6%	0.0%	18.0	47.9
2	v3_f16	v64p_paired_owner	raw	v1	187/205	91.2%	91.7%	94.6%	0.0%	20.1	51.8
3	v3_f32	v20_aux	raw	v1	186/205	90.7%	93.7%	94.6%	0.0%	12.7	30.7
4	v3_f16	v20_aux	raw	v1	186/205	90.7%	93.7%	94.6%	0.0%	14.0	31.6
5	v2_f32	v64p_paired_owner	raw	v1	186/205	90.7%	91.7%	93.7%	0.0%	27.3	63.5
6	v2_f16	v64p_paired_owner	raw	v1	186/205	90.7%	82.6%	93.7%	50.0%	63.0	105.3
7	v2_f16	v20_aux	raw	v1	185/205	90.2%	93.9%	93.7%	50.0%	81.9	90.1
8	v2_f32	v20_aux	raw	v1	185/205	90.2%	93.7%	93.7%	0.0%	19.2	50.8
9	v3_f32	v19	raw	v1	185/205	90.2%	92.7%	94.6%	0.0%	6.1	14.2
10	v3_f16	v19	raw	v1	185/205	90.2%	92.7%	94.6%	0.0%	7.5	18.2
11	v3_f32	v64c_stochmel	raw	v1	185/205	90.2%	90.7%	94.6%	18.8%	16.1	42.0
12	v3_f16	v64c_stochmel	raw	v1	185/205	90.2%	90.7%	94.6%	18.8%	18.8	46.1
13	v2_f32	v64c_stochmel	raw	v1	185/205	90.2%	90.7%	93.7%	18.8%	28.7	72.2
14	v3_f32	v64c_stochmel	raw	v2	185/205	90.2%	90.2%	94.6%	25.0%	11.2	25.2
15	v3_f32	v64p_paired_owner	raw	v2	185/205	90.2%	90.2%	94.6%	6.2%	11.9	26.5
16	v3_f16	v64c_stochmel	raw	v2	185/205	90.2%	90.2%	94.6%	25.0%	12.0	26.9
17	v3_f16	v64p_paired_owner	raw	v2	185/205	90.2%	90.2%	94.6%	6.2%	12.3	27.1
18	v3_f32	v64_paired_negatives	raw	v1	185/205	90.2%	90.2%	94.6%	6.2%	21.9	58.9

Sources: recordings/results (2026-10-24 batch) | ME2 holdout 202 clips (86 real + 100 synth + 16 OOS) | owner takes.